Epistemic Blinding: An Inference-Time Protocol for Auditing Prior Contamination in LLM-Assisted Analysis

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Abstract

We built an agentic system that uses large language models to reason across multiple biological datasets for drug target prioritization. During development, we discovered that the LLM's outputs silently blend data-driven inference with memorized priors about named entities—and the blend is invisible: there is no way to determine, from a single output, how much came from the data on the page and how much came from the model's training memory. To address this, we introduce *epistemic blinding*, a simple inference-time protocol that replaces entity identifiers with anonymous codes before prompting, then compares outputs against an unblinded control. The protocol does not make LLM reasoning deterministic, but it restores one critical axis of auditability: measuring how much of an output came from the supplied data versus the model's parametric knowledge. In our original domain—oncology drug target prioritization across four cancer types—blinding changes 16% of top-20 predictions while preserving identical recovery of validated targets. The known biology is the same, but the discovery frontier shifts from literature-derived to data-driven nominations. Recognizing that this problem is not specific to biology, we generalized the protocol into an open-source, model-agnostic tool and tested it on S&P 500 equity screening, where brand-recognition bias reshapes 30–40% of top-20 rankings across five random seeds. We do not claim that blinded analysis produces better results. We claim that without blinding, there is no way to know to what degree the agent is adhering to the analytical process you designed.

1 Introduction

Consider a concrete example. An LLM is given a table of 100 genes with quantitative features for colorectal cancer and asked to rank the top 20 drug targets. When the gene names are visible, the model ranks KRAS at position #1 and justifies the choice by citing "proven therapeutic tractability via covalent RAS inhibitors." When the identical features are presented with anonymous labels, the gene corresponding to KRAS—now labeled Gene_088—is ranked #5, below genes with higher mutation frequencies and stronger convergence signals. The phrase "proven therapeutic tractability via covalent RAS inhibitors" does not appear anywhere in the provided data. It comes from the model's training corpus.

This is not a failure of the model. It is a failure of the experimental design. The model was asked to perform data-driven inference but was simultaneously given access to its parametric knowledge about the entities in question. The two signals—data and memory—are entangled in the output with no means to distinguish them.

We call this *parametric knowledge contamination* and introduce *epistemic blinding* as a protocol to detect and measure it. The idea is conceptually identical to blinding in clinical trials: prevent the analyst from accessing information that could bias the analysis. Here, the analyst is an LLM, and the blinding is achieved through string replacement at the prompt level.

Why this matters beyond accuracy. The natural question is whether blinded or unblinded analysis produces "better" results. We deliberately resist this framing. In some cases, training priors are genuinely informative—a model that knows KRAS is druggable is incorporating real-world knowledge that might improve a recommendation. In other cases, those same priors mask data-driven candidates that could represent novel biology. The point of epistemic blinding is not to suppress the model's knowledge. It is to make the *influence* of that knowledge visible and measurable—to restore an axis of auditability that is otherwise absent.

When we write traditional software, every step is traceable. We can follow a data point from input through every transformation to output and verify that the logic matches our intent. LLM-based analysis breaks this: the model's reasoning is a black box in which supplied data and memorized knowledge are mixed without attribution. Epistemic blinding does not make agentic reasoning deterministic. But it brings back something we expect from deterministic code: the ability to audit whether the tool is operating on the inputs we gave it, or on something else entirely.

Contributions. This work originated in a specific need—building an agentic system for multi-dataset biological reasoning without prior contamination—and generalized into a domain-agnostic protocol. We present (1) a formal description of epistemic blinding, including treatment of cross-dataset consistency and subtle leak sources; (2) an open-source, config-driven implementation that works with any LLM; (3) a controlled experiment in oncology drug target prioritization, the domain that motivated the work, showing systematic fame bias across four cancer types; and (4) a replication in S&P 500 equity screening demonstrating that the same phenomenon—and the same fix—transfers across domains.

2 Related Work

Data contamination in LLM evaluation. The problem of training data leaking into benchmark evaluations is well-studied [1–3]. Our work addresses a related but distinct problem: the model's *general knowledge* contaminating *domain-specific reasoning*, not memorized test sets inflating metrics.

Entity bias in language models. Wang et al. [4] demonstrated entity-specific biases in factual question answering and proposed causal interventions that perturb entity names. We extend this from factual QA to scientific and financial analysis, where the consequences extend beyond accuracy to resource allocation and discovery.

Anonymization in multi-agent reasoning. Choi et al. [5] showed that anonymizing agent identities in multi-LLM debate reduces conformity bias. We extend this principle from social identity to

epistemic identity—anonymizing the *objects of reasoning* rather than the *reasoning agents*.

Pretraining leakage in protein language models. Hermann et al. [6] demonstrated that protein language model pretraining creates data leakage that distorts downstream task performance by 11.1%. Their solution—pretraining-aware data splits—addresses the problem at the data level. Ours addresses it at inference time, requiring no retraining.

LLMs for biological analysis. Hu et al. [7] used GPT-4 for gene set function discovery with blinded *human* evaluation of the LLM's outputs. Notably, they did not blind the LLM itself—the model received real gene names and could draw on its training knowledge. Our work completes this design by blinding both sides.

3 Method

3.1 Protocol

Epistemic blinding is an inference-time intervention requiring no model modification. Given one or more datasets containing named entities:

(1) **Identify entity columns.** Determine which columns contain identifiers that may have uneven representation in the LLM's training corpus (gene symbols, tickers, drug names, team names, author names).
(2) **Build a shared mapping.** For each entity type, collect every distinct value across all datasets. Shuffle deterministically with a stable seed and assign anonymous codes of the form `{Prefix}_{number}` (e.g., Gene_001, Company_042). The same entity receives the same code everywhere it appears, preserving cross-dataset reasoning.
(3) **Identify and mitigate subtle leak sources.** Beyond the entity name itself, correlated features may reveal identity. Market capitalization and sector together can identify most mega-cap stocks; a 42% mutation frequency in colorectal cancer effectively identifies KRAS. We distinguish *obvious* leak sources (the entity name) from *subtle* ones (features that correlate strongly enough with identity to partially unblind the model). The protocol requires the analyst to enumerate potential subtle leaks and decide whether to normalize, bin, or drop them.
(4) **Shuffle and render.** Randomize row order with a fixed seed and format the anonymized data as a prompt, preserving column headers, units, and feature descriptions.
(5) **Run A/B comparison.** Send the blinded prompt to an LLM in a fresh session with no prior context. Separately, send the matched unblinded prompt to the same model in an independent session. Compare outputs using set overlap, rank shift, and qualitative justification analysis.
(6) **De-anonymize.** Resolve anonymous codes back to real identifiers using the mapping table.

The protocol is illustrated in Figure 1.

3.2 What Gets Blinded and What Doesn't

The goal is to blind information that activates training priors while preserving information the model legitimately needs for reasoning. Entity identifiers and proper nouns with a training-data footprint are blinded. Feature column names, units, numeric values, categorical labels needed for task context, dates, and data source names are preserved. Some categories sit in the middle—disease names, for instance, carry priors but also anchor the analytical task. The protocol supports partial blinding via per-column configuration.

3.3 Implementation

We release an open-source, config-driven tool[1] comprising three scripts: `blind.py` (reads a YAML config, produces matched blinded/unblinded prompts plus a mapping file), `deblind.py` (reverses the mapping in a model response), and `compare.py` (computes A/B metrics: set overlap, Jaccard index, mean rank delta, Kendall τ, and optional fame-bias statistics). The tool is model-agnostic and has also been implemented as a Claude Code skill for one-command blinding within agentic workflows.

4 Case Study: Oncology Drug Target Prioritization

4.1 Task and Data

We tested epistemic blinding on the task of ranking the top 20 most promising drug targets for a given cancer indication, given quantitative features derived from patient tumor sequencing and experimental biology.

The convergence hypothesis. Our features are derived from a biological premise: a gene is a promising drug target when independent experimental evidence converges on the same molecular neighborhood. Concretely, if a gene's protein structure neighbors (ESM2 [8]), transcriptomic co-expression partners (Geneformer [9]), and functional domain relatives (ProtT5 [10]) are all enriched for somatically mutated genes in a given cancer, that gene sits at a biological hub for that disease.

Data sources. Somatic mutation data from the cBioPortal TCGA PanCancer Atlas [12] (~1M mutations, 14 cancer studies) at molecular subtype resolution. Experimental neighbor graphs from three independent foundation models trained by different groups on different data modalities. Genetic constraint from gnomAD v4.1 pLI scores.

Feature set. Each gene is characterized by seven numerical features: somatic mutation frequency, binary mutation significance, pLI (loss-of-function intolerance), and enrichment scores from ESM2, Geneformer, and ProtT5 neighbor graphs, plus total neighbor count. These features are purely quantitative—a gene's enrichment of 15.4 carries the same information whether labeled SCN1A or Gene_016.

Unbiased data assembly. Literature-mined gene–disease relationships, pre-annotated single-cell atlases with canonical marker genes, and expert-curated pathway databases were deliberately excluded. Each included source was selected for a single property: it measures biology without requiring prior knowledge of what the answer should be. Every source was accompanied by a context card [11].

[1] https://github.com/mcuccarese/epistemic-blinding

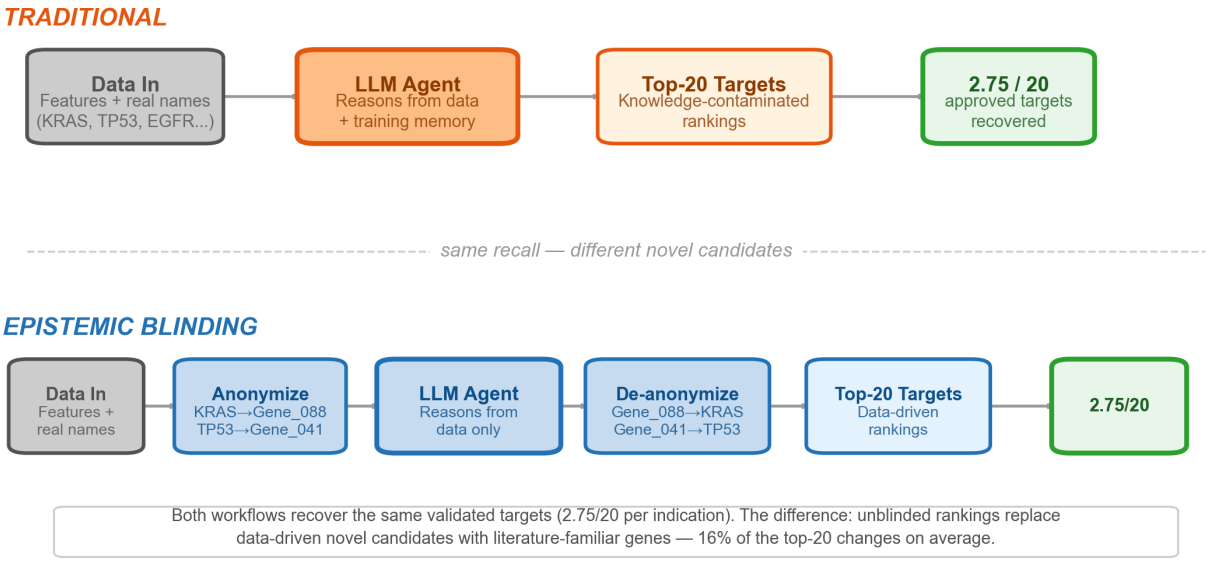

Figure 1: Traditional vs. epistemic blinding workflow. Both approaches receive identical quantitative features and recover identical numbers of validated drug targets. The critical difference is in which novel candidates are surfaced: the unblinded workflow promotes literature-familiar genes while the blinded workflow ranks purely on feature strength.

4.2 Experimental Design

For each of four oncology indications spanning a range of feature signal strength, we selected the top 100 candidate genes using a deterministic scoring function that operates solely on the numerical features—no gene names or identities enter the selection. Matched blinded and unblinded prompts were presented to Claude (Anthropic) in fresh sessions.

Indication	Sig. genes	Targets	Signal
Acute myeloid leukemia	30	6	Strong
Pancreatic adenocarcinoma	32	1	Concentrated
Chromosomally unstable CRC	300	1	Moderate
IDH-wildtype glioblastoma	110	3	Weak

Table 1: Indications tested, spanning a range of feature signal strength.

4.3 Results

Aggregate. Average set overlap between blinded and unblinded top-20 lists was 84%, meaning blinding changes 16% of predictions. Target recovery was identical in both conditions (average 2.75 approved targets per indication). Blinding does not degrade recovery of validated biology—it changes which *novel* candidates are nominated.

	Overlap	Only B	Only U	Tgt
AML	90%	2	2	6/6
Pancreatic	80%	4	4	1/1
CRC	90%	2	2	1/1
GBM	75%	5	5	3/3
Average	**84%**	3.25	3.25	2.75

Table 2: Set overlap between blinded and unblinded top-20 lists. Target recovery (rightmost column) is identical in both conditions.

Fame bias. The direction of change is systematic (Figure 2). Well-known genes are promoted when named, and obscure genes with strong features are demoted. PTEN jumped from #15 (blinded) to #3 (unblinded) in GBM, with the model citing "the defining PI3K/AKT pathway tumor suppressor." DPP8, which had the strongest convergence signal in the entire GBM gene set, dropped from #3 to #9—displaced by genes the model recognized as established GBM biology despite their weaker features. KRAS was promoted from #5 to #1 in CRC despite another gene having twice the mutation frequency.

Contamination scales with feature ambiguity. AML (strong signal, few drivers) shows 90% overlap. GBM (weak signal, many genes at similar low frequencies) shows 75% overlap. When features clearly

differentiate candidates, the model follows the data. When features are ambiguous, the model fills the gap with training knowledge.

4.4 The Smoking Gun

The most direct evidence of contamination comes from the model's own justifications (Table 3).

5 Case Study: S&P 500 Value Screening

To test generalizability, we applied the protocol to equity analysis. An LLM was asked to rank the 20 most attractive value investments from ~500 S&P 500 companies, given four layers of fundamentals: valuation (P/E, P/S, P/B, EV/EBITDA, PEG, FCF yield), growth (revenue, earnings), quality (margins, ROE, ROA, leverage), and shareholder returns (dividend yield). Because market capitalization and sector together can identify most mega-cap stocks, all features were normalized. Ticker symbols were the only entity column blinded.

We ran five independent seeds (42, 137, 256, 389, 501). Results:

On average, 7 of the top 20 change when tickers are revealed—a 35% reshaping of the recommendation. The effect is systematic (Figure 3): tickers like ELV and CI were promoted in 4 of 5 runs; CTRA was demoted in 3 of 5 runs. The model is not randomly noisy—it is consistently biased by brand recognition.

6 Discussion

6.1 Auditability, Not Accuracy

We want to be explicit about what epistemic blinding does and does not provide. It provides *auditability*: the ability to measure how much of an LLM's output came from the supplied data versus its training memory. It does not provide *accuracy*: we make no claim that blinded results are more correct.

In some cases, training priors will be informative. In other cases, they will be misleading. The analyst needs to know which situation they are in, and they cannot know that from an unblinded analysis alone.

This framing is analogous to how we think about deterministic code. When a traditional program produces an unexpected output, we can trace the execution path. When an LLM-based agent produces an unexpected output, we cannot. Epistemic blinding does not make the agent deterministic—but it restores one specific axis of traceability: did the output come from my data, or from somewhere else?

6.2 When to Blind

A practical heuristic: if you would blind a human analyst performing the same task, you should blind the LLM. Three conditions should hold: (1) the data contains decision-relevant signal, (2) the entities have uneven representation in the training corpus, and (3) someone will act on the output.

6.3 What This Does Not Fix

Sampling bias. If certain signals are over-represented in data collection, they will be over-represented regardless of blinding. In our oncology case study, we addressed this by deliberately excluding literature-mined gene–disease associations.

Structural leakage. Some feature combinations reveal identity even without the name. Mitigations include normalizing features, binning values, or dropping identifying columns.

Semantically meaningful names. In some domains, names encode information (IUPAC chemical names, drug suffixes). Blinding these destroys legitimate signal. The protocol should be applied to identity, not to functional nomenclature.

7 Limitations

(1) **Single model.** All experiments used Claude (Anthropic). The magnitude of contamination may differ across models.
(2) **Binary comparison.** We compared fully blinded vs. fully unblinded. Partial blinding is unexplored.
(3) **No ground truth for novel candidates.** We can measure that the discovery frontier shifts but cannot determine which frontier is "better" without prospective validation.
(4) **Run-to-run variance.** The S&P 500 experiment used five seeds; the oncology experiment used single runs per indication.

8 Conclusion

Revealing entity identifiers to an LLM during data-driven analysis systematically biases its outputs toward well-known entities and away from conclusions supported by the data alone. The intervention to detect this—epistemic blinding—is trivially simple, requires no model modification, and preserves the model's analytical reasoning capabilities.

In oncology, blinding changed 16% of top-20 drug target predictions while maintaining identical recovery of validated targets. In equity analysis, blinding changed 35% of top-20 value rankings. In both domains, the direction of bias was systematic: famous entities were promoted, obscure entities with strong features were demoted.

We propose that epistemic blinding should become standard practice whenever LLMs are used for data-driven analysis of named entities. Not because blinded results are necessarily better—but because without blinding, the analyst has no way to know whether the agent is reasoning from the data they provided or from something it memorized long before the conversation began.

Code availability. The epistemic blinding tool, including configuration schemas, worked examples, and comparison scripts, is available at https://github.com/mcuccarese/epistemic-blinding.

References

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[8] Z. Lin et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379:1123–1130, 2023.

	Blinded (Gene_088, ranked #5)	Unblinded (KRAS, ranked #1)
Opening	"Gene_088 ranks #5 based on quantitative mutation features"	"KRAS ranks #1 as the most actionable target"
Shared reasoning	*Both cite: high mutation frequency (38%), recurrent codon-12 hotspot, strong statistical enrichment, exclusively missense mutations*	
Additional reasoning	"GTPase domain concentrated mutations suggest gain-of-function mechanism"	**"Proven therapeutic tractability via covalent RAS inhibitors (sotorasib, adagrasib)"** **"Central node in RAS/MAPK pathway with extensive clinical validation"**
Conclusion	"Exceptionally high recurrence at a single hotspot with strong functional convergence"	"KRAS is the defining oncogenic driver in CRC with validated druggability and multiple approved therapies"

Table 3: Side-by-side LLM justifications for the same gene (KRAS / Gene_088) in colorectal cancer. The highlighted phrases in the unblinded column are not derivable from any provided feature—they come from the model's parametric knowledge. Same gene, same features, different rank (#5 vs. #1), different reasoning.

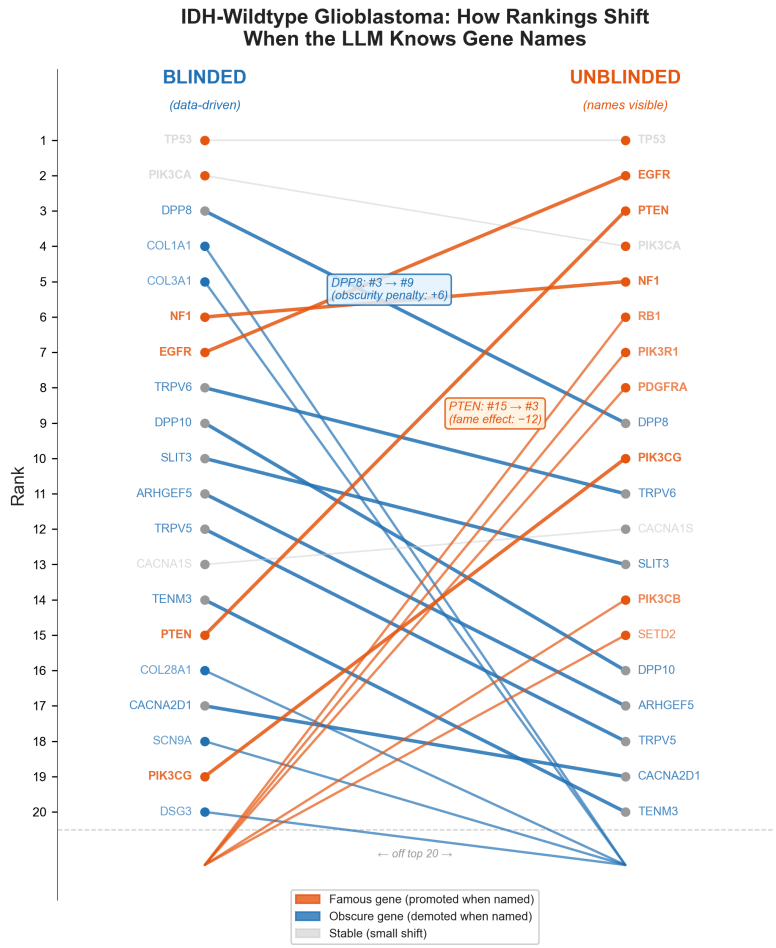

Figure 2: Rank-shift slope chart for IDH-wildtype glioblastoma. Lines connect the same gene's rank in the blinded (left) and unblinded (right) conditions. Orange: famous genes promoted when named. Blue: obscure genes with strong features demoted when named.

[9] C. V. Theodoris et al. Transfer learning enables predictions in network biology. *Nature*, 618:616–624, 2023.

[10] A. Elnaggar et al. ProtTrans: Toward understanding the language of life through self-supervised learning. *IEEE TPAMI*, 44:7112–7127, 2022.

Metric	Mean ± SD	Range
Top-20 overlap	13.0 ± 1.0	12–14
Jaccard index	0.48 ± 0.06	0.43–0.54
Mean rank delta	3.1 ± 1.1	1.9–4.4
Kendall τ	0.52 ± 0.17	0.28–0.67

Table 4: S&P 500 blinded vs. unblinded comparison across five seeds.

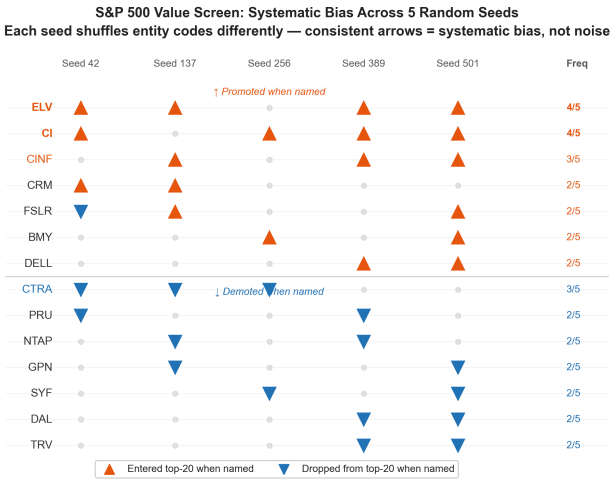

Figure 3: S&P 500 consistency across five random seeds. Each seed shuffles entity codes differently. Consistent arrows indicate systematic bias, not stochastic noise.

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